SOFTWARE

REQUIREMENT

AND

SPECIFICATION

(SRS)

2. SOFTWARE REQUIREMENT AND SPECIFICATION (SRS)

1) Introduction:

Edu-Connect is a web-based Learning Management System (LMS) designed to improve communication and resource sharing between students, teachers, and administrators. It allows teachers to upload study materials, share educational videos, and send notifications, while students can access learning resources, provide feedback, and interact with content. The system is developed using PHP and MySQL and runs on a local server using XAMPP. It provides a centralized environment that supports efficient learning and academic collaboration.

1.1 Overall Description:

Edu-Connect is a standalone web application for students, teachers, and administrators in an educational environment. Teachers can upload study materials, share educational videos, and broadcast notifications. Students can browse learning resources, comment, provide feedback, and search for other users.

The system uses PHP, MySQL, HTML, CSS, and JavaScript, and operates locally on XAMPP. It creates a centralized digital space for academic collaboration. The platform supports three main user roles:

- Students
- Teachers
- Administrators

Each role has specific permissions and features.

1.2 Product Perspective:

Edu-Connect is developed as a standalone web application running on a local server. It is independent of any other LMS platform and is built from scratch without using frameworks.

The system consists of:

- Web Browser Interface – Used by students, teachers, and administrators.
- Application Layer – Handles the business logic and user actions.
- Database Layer – Stores all user data, materials, videos, and notifications.

Edu-Connect can later be extended for remote servers, external tool integration, or mobile access.

1.3 Architecture:

Edu-Connect follows a three-layer architecture for maintainability and scalability.

Data Layer: The Data Layer manages all database operations. It uses MySQL, accessed through MySQL in PHP. It includes tables for users, students, teachers, materials, notifications, study videos, likes, comments, and feedback. The file db.php provides the database connection.

Application Layer: The Application Layer contains the system's core logic. It processes user requests, handles authentication, performs database operations, and manages tasks like uploading materials, posting videos, sending notifications, and collecting feedback.

Presentation Layer: The Presentation Layer is the user interface. It includes dashboards, forms, and content pages, implemented using HTML, CSS, JavaScript, and PHP templates. It provides different interfaces for students, teachers, and administrators with a modern design system.

1.4 Product Features

1. User Authentication: Users can register and log in securely. Administrator approval is required before access.

2. Role-Based Access Control: Each user role has different permissions **(**Students, Teachers, Administrators).

3. Study Material Management: Teachers can upload PDFs, videos, images, or external links. Students can browse and download them. Administrators can view or delete any material.

4. Study Videos Platform: Users can share YouTube videos, which are embedded automatically. Features include posting videos, liking, saving, and commenting. Administrators can delete videos.

5. Notification System: Teachers and administrators can send notifications to students or all users. Notifications appear on dashboards with unread counts and can be marked as read.

6. User Search and Profiles: Users can search for students or teachers by name, ID, or college. Public profiles display basic details and profile photos.

7. Feedback System: Users can give a 1–5 star rating with optional comments. Each user can submit one feedback but can update it later.

8. Administrative Controls: Administrators can approve or reject user registrations, monitor uploaded materials, manage notifications, and view platform statistics.

2) User Characteristics:

The Edu-Connect system is designed for students, teachers, administrators, and educational institutions.

- Students – Access study materials, videos, notifications, and give feedback. Basic computer and internet skills are enough.
- Teachers – Upload learning resources, videos, and notifications. Moderate web skills are needed.
- Administrators – Manage user approvals, monitor uploads, manage notifications, and view statistics.
- Educational Institutions – Schools, colleges, and training centers can use Edu-Connect to support digital learning and collaboration.

3) General Constraints:

The development and operations of **Edu-connect** are subjected to certain limitations

1. **Technical Constraints** – Must run using PHP and MySQL on XAMPP; requires a stable browser and server connection.
2. **Platform Constraints** – Web-based; must be accessed via computers or laptops with internet and browsers.
3. **Resource Constraints** – Limited storage, server performance, and file upload size.
4. **Regulatory Constraints** – Must follow privacy, content, and academic policies.
5. **User Constraints** – Users need basic computer knowledge; role-based access; administrator approval required.

4) Assumptions:

Several assumptions are made during apps development:

- Users have basic computer and web browser knowledge.
- The system runs on XAMPP (PHP + MySQL).
- Users have stable internet connections.
- Teachers upload valid educational materials.
- Administrators monitor and approve registrations.
- Server and database have sufficient storage.

5) Special Requirements :

- File Uploads – Support PDFs (up to 50MB), videos, images, and links in designated folders.
- Role-Based Access – Different features per user role.
- Real-Time Notifications – Unread counts and AJAX mark-read functionality.
- Video Management – Embed YouTube videos; like, save, and comment asynchronously.
- Feedback System – One rating per user, updatable.
- Security – Passwords hashed with bcrypt; inputs sanitized.
- Browser Compatibility – Works on Chrome, Firefox, Edge.
- Extensibility – Architecture supports future features like mobile support or new file types.

5.1 **Software Requirements:**

The system requires:

- OS – Windows 10 or higher.
- Server – XAMPP (Apache + MySQL).
- Backend – PHP 8+.
- Database – MySQL via MySQLi.
- Frontend – HTML, CSS, JavaScript.
- Browser – Chrome, Firefox, or Edge.

5.2 Hardware Requirements:

- Processor – Minimum Intel Core i3; recommended Intel Core i5 or higher.
- Memory (RAM) – Minimum 4 GB; recommended 8 GB or more.
- Storage – Minimum 100 GB; recommended 250 GB SSD or higher.
- Display – Minimum 1366 × 768 pixels.
- Server (XAMPP) – Dual-core CPU, 8 GB RAM, 250–500 GB storage, LAN/Wi-Fi network.

6) Functional Requirements

- **User Registration & Login** – Users can register and log in; admin approval required.
- **Role-Based Access** – Features restricted based on user role.

- **Study Materials** – Upload, browse, download materials; admin can delete.
- **Study Videos** – Post, like, save, comment; admin can delete.
- **Notifications** – Send, view, mark as read; unread badges displayed.
- **User Search & Profiles** – Search by name, ID, or college; public profile visible.
- **Feedback** – 1–5 star ratings with comments; updatable.
- **Admin Controls** – Approve/reject users, monitor materials and notifications, view statistics.

6.1 Design Constraints

- Technology – PHP + MySQL; frontend with HTML, CSS, JS only.
- Server – Runs on XAMPP; works on Chrome, Firefox, Edge.
- Architecture – Three layers: Data, Application, Presentation.
- UI – Must use layout system; follow CSS design; display on standard desktops/laptops.
- File Uploads – Stored in correct folders; follow allowed types and sizes.
- Security – Inputs sanitized; passwords hashed.
- Roles – Users access only permitted features; actions like delete depend on role.

6.2 System Attributes:

- **Reliability** – System should run without crashes; data saved correctly.
- **Availability** – Accessible whenever server is running.
- **Performance** – Pages load quickly; multiple users supported.
- **Security** – Accounts protected; unauthorized access prevented.
- **Maintainability** – Easy to update or fix; organized code layers.
- **Scalability** – Supports more users, materials, videos; easy to add features.
- **Usability** – Easy to navigate; simple interaction with notifications, videos, and materials.

6.3 Other Requirements

- **Backup & Recovery** – Regular database backups; data recoverable after crashes.

- **Legal & Regulatory** – Follow privacy rules; only educational content allowed.
- **Compatibility** – Works on modern browsers; runs on standard desktops/laptops.
- **Localization** – English used; other languages may be added later.
- **Documentation** – Simple user guides; technical documentation for admins.
- **Extensibility** – Future features like mobile support or new file types can be added